Young 2018 Kidney (Wilms / RCC) — summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/25 samples (≥100 cells) intermix on one continuous manifold (kNN = 1 component; 0.91 of the perfect-mixing ceiling), no per-patient islands (radius-graph: 3 specks stripped). No proliferation compartment (most-cycling lobe 1.5%); module means near-identical across all 3 lobes → whole manifold.

Gate 2 — module geography → GO (non-branching)

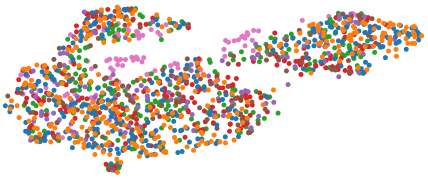
Single dominant pSom→Crest axis: pSom high over the body declines along pseudotime (ρ=−0.75); Crest anti-correlated (r=−0.78), rising to a terminal readout (ρ=+0.72; Crest-high decile mean pt=0.89). Gast weakly structured (ρ=+0.39). Joint pSom&Crest-high depleted (enrichment 0.22) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

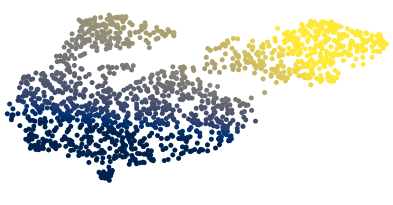
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

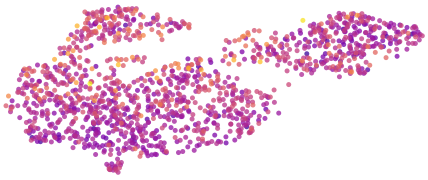
Gate 1: integration
Kidney malignant — samples



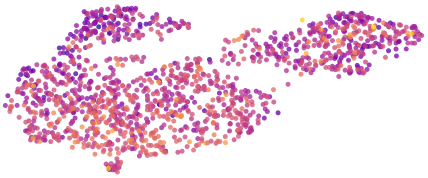
Trajectory / root
Pseudotime



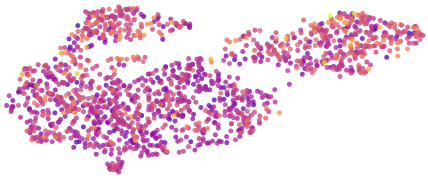
Gast
Gastrulation Signature



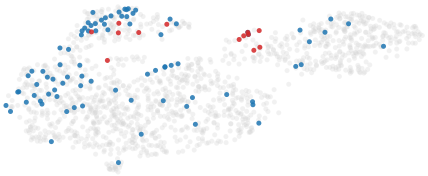
pSom
Somitogenesis Signature



Crest
Neural Crest Signature



EMT-TFs
EMT-TF Expression ZEB2 (early) · FOXC1 (late)



Dataset. Wilms tumour / RCC (Young et al., 2018), 10x — 66 samples / 125,139 cells (master scope); malignant only: 2,055 cells → 1,438 after ≥100/sample threshold (7/25 samples) → 1,435 after island removal enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Young's kidney malignant manifold is a single monotone somitogenesis→terminal-neural-crest gradient (pSom-high body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The manifold is continuous and cell-cycle-free (most-cycling lobe 1.5%, Gast vs g2m r=−0.02), so no proliferation caveat; the whole manifold is kept (Bi/Steele no-scoping precedent). Module means Gast −0.008 / pSom +0.042 / Crest −0.029 (pSom-dominant → Linear-pSom/Crest). EMT-TFs (3% detection floor, FOXC/PRRX families included, VIM excluded): ZEB2 (early/root, 23.3% expressing, ρ=−0.20) ↔ FOXC1 (late/terminus, 3.6%, ρ=+0.03) — FOXC1 surfaces only via the widened panel; end-separation weak (Δpt=0.27). Curve fold-free (self-approach 2.84, 0 forks).

Parameter choices

Min cells/sample	≥100 floor (7/25; gap non-dominant)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = pSom-high, Crest-low	**Overdispersed**	262 genes; n_pcs=30
Geometry	Linear-pSom/Crest (non-branching)	**EMT-TFs**	ZEB2 (early) · FOXC1 (late)